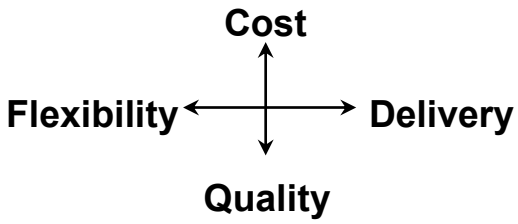
ISyE 315 DISCUSSION 1 (Chapters 1-2)

Definitions/Concepts	
Operations Management	The design, operation, and improvement of the systems that create and deliver the firm's primary products and services. "…creating great value to the customer while reducing the cost of delivering the good or service."
Transformations	Using resources to convert inputs into some desired outputs 6 Transformations: Physical, locational, exchange, storage, physiological, informational
Sustainability	The ability to meet current resource needs without compromising the ability of future generations to meet their needs *Triple Bottom Line*: Economic Prosperity, Environmental Stewardship, Social Responsibility
Core Capabilities / Competitive Dimensions	Skills that differentiate a manufacturing or service firm from its competitors • *Cost*: make it cheap • *Product Quality/Reliability*: make it good • *Delivery Speed (lead time):* make it fast • *Delivery Reliability*: deliver what/when promised • *Cope with changes in demand*: change volume • *Flexibility/new product introduction speed*: change it • *Product specific criteria*: support it
Trade-offs	 Understand the desire for balance: efficiency, effectiveness, value
Order Qualifiers	Set of dimensions used to screen a product or service as a candidate for purchase
Order Winners	Set of dimensions that differentiate the product or services of one firm from those of another. *Order winners can become order qualifiers over time
Risk Mitigation	Identify potential sources → Assess risks → Develop mitigation/correction plan Severity x Occurrence x Detection
KPI	Key Performance Indicators- need to measure in order to know what/where to improve "*Balanced Scorecard*" is desirable Ex: productivity, utilization, lead time, etc.
Productivity	A *relative* measure of how well resources are used (can also be partial or multifactor) Ex: Labor productivity can be looked at as net income or revenue per employee.

Equations	
Productivity	$Productivity = \frac{Output}{Input} = \frac{Goods\ and\ Service\ Produced}{Resources\ Used}$
Asset Productivity	$Receivable\ turnover\ = \frac{Annual\ Credit\ Sales}{Average\ Accounts\ Receivables}$ $Inventory\ turnover\ ratio = \frac{Cost\ of\ Goods\ Sold}{Average\ Inventory\ Value}$ $Asset\ turnover\ ratio = \frac{Revenue\ (or\ Sales)}{Total\ Assets}$
Input	$Labor\ cost\ +\ Material\ cost\ +\ Overhead\ costs$
Output	$Made\ Volume\ *\ Cost/unit$
Percent Change	$Percent\ Change = \frac{New - Old}{Old} * 100$

Example Problems

1. A clothing retailer's 2022 financial data are listed below. Calculate receivable turnover ratio, inventory turnover ratio, and asset turnover ratio.

Gross Income	$40,560,000
Total Sales	37,620,000
Total Credit Sales	33,472,000
Net Income	5,075,000
Cost of Goods Sold	23,360,000
Total Assets	19,580,000
Average Inventory	4,575,000
Average Receivables	6,395,000

$$Receivable\ turnover\ = \frac{Annual\ Credit\ Sales}{Average\ Accounts\ Receivables} = \frac{\$33{,}472{,}000}{\$6{,}395{,}000} = \mathbf{5.234}$$

$$Inventory\ turnover\ ratio = \frac{Cost\ of\ Goods\ Sold}{Average\ Inventory\ Value} = \frac{\$23{,}360{,}000}{\$4{,}575{,}000} = \mathbf{5.11}$$

$$Asset\ turnover\ ratio = \frac{Revenue\ (or\ Sales)}{Total\ Assets} = \frac{\$37{,}620{,}000}{\$19{,}580{,}000} = \mathbf{1.92}$$

2. A team of workers made 400 #BeatPurdue t-shirts for the upcoming football game (Sep 22nd), each of which is valued by its standard cost of $10 each (before markups for other expenses and profit). The accounting department reported that for this job the actual costs were $400 for labor, $1000 for materials, and $300 for overhead. What is the overall productivity? If the minimum wage were to increase, what would happen to the overall productivity?

a) 2.85, increase b) 2.35, increase c) 2.85, decrease **d) 2.35, decrease**

$$Input = Labor\ cost\ +\ Material\ cost\ +\ Overhead\ costs = \$1000 + \$400 + \$300 = \mathbf{\$1700}$$

$$Ouput = Made\ Volume\ * \frac{Cost}{unit} = \$400 * \$10 = \mathbf{\$4000}$$

$$Productivity = \frac{Output}{Input} = \frac{\$4000}{\$1700} = \mathbf{2.35}$$

If minimum wage increases, labor cost increases so the denominator increases, **causing productivity to decrease**.

3. Referring to Problem 2 - some changes were made and for the Rutgers game (Oct 7th)… accounting reported that the expenses for the same amount of units at the same cost were $350 for labor, $800 for materials, and $325 for overhead costs. Will this result in increased or decreased productivity? What is the percent change?

a) Decrease; 15.4% **b) Increase; 15.4%** c) Increase; 13.3% d) Decrease; 13.3%

$$Input = Labor\ cost\ +\ Material\ cost\ +\ Overhead\ costs = \$800 + \$325 + \$350 = \$\mathbf{1475}$$

$$New\ Productivity = \frac{Output}{Input} = \frac{\$4000}{\$1475} = \mathbf{2.71}$$

$$Percent\ Change\ Compared\ to\ Old\ productivity = \frac{New - Old}{Old} * 100 = \frac{2.71 - 2.35}{2.35} * 100 = \mathbf{15.32}\%$$

Productivity increased by 15.32%

4. Margo and John are baking cakes for different events in their new job. In one week (20 hrs each) they baked a total of 132 cakes, 12 of which were tiered cakes. Tiered cakes are sold for $50 each whereas the standard ones sell for $35. What is the overall labor productivity ratio?

a) 40 hrs b) $12/hr c) $131/hr **d) $120/hr**

$$Productivity = \frac{Goods\ and\ Service\ Produced}{Resources\ Used} = \frac{(\$50 * 12) + (\$35 * 120)}{40\ hours} = \frac{\$4800}{40} = \$\mathbf{120/hour}$$